

COURSE SYLLABUS

Course title: Digital RTL Design and Verification with SystemVerilog

1. Course title:	Digital RTL Design and Verification with SystemVerilog
2. Course code:	
3. Credits:	04
4. Course plan	
- Lecture:	3.5 credits (1 credit ~ 15 hours)
- Laboratory:	1.5 credits (1 credit~30 hours)
5. Instructors	Tai Le is a seasoned expert in Electronic Design Automation (EDA) and semiconductor design, with over 30 years of global experience spanning ASIC, custom, and mixed-signal design flows. His career includes pivotal roles in software development and CAD management at major design centers across the US, Europe, and Asia. He has held senior technical positions at leading firms such as Micron, Hitachi America, and Avant! (now part of Synopsys). Education: - M.S. in Computer Engineering, Santa Clara University - B.S. in Computer Science and Computer Mathematics, Cum Laude, California State University, Northridge
6. Prerequisite requirements:	Knowledge: GPA: >=2.8 Have studied or knowledge of the following modules: - Principles of Computer Programming - Fundamentals of Electrical Engineering - Electric Circuits - Digital Circuits and Systems - Basic hands-on experience with UNIX OS environment English: Must be at level 3/6 according to Vietnam's 6-level foreign language proficiency framework
7. Course type:	☐ Required ☒ Selected Elective ☐ Free elective
8. Knowledge Cluster	☐ Math and basic science ☐ General education ☒ Major education ☐ Internship/Capstone Project

9. Course description:

This course introduces students to the core principles of digital hardware design using SystemVerilog, covering everything from basic syntax and simulation to synthesizable RTL modeling. Through hands-on coding and simulation exercises, learners will gain the skills to describe, verify, and implement digital logic systems. The course also provides insight into industry-standard design flows and prepares students for real-world IC design challenges.

10. Course learning outcomes (CLOs):

At the end of this course, students should be able to:

No	(CLOs)
1.	Explain the purpose of HDLs/VDLs in the IC development process
2.	Describe the major steps in a typical ASIC/SOC design and verification flow
3.	Understand SystemVerilog syntax, data types, and operators
4.	Write simple SystemVerilog modules with ports and parameters
5.	Use procedural blocks and control flow statements for simulation modeling
6.	Understand the fundamentals of event-driven simulation
7.	Write tasks and functions, use system tasks and functions
8.	Explain the role of SystemVerilog simulation models within an ASIC library context.
9.	Model combinational and sequential standard cells using SystemVerilog constructs.
10.	Learn RTL design principles and styles
11.	Able to write RTL models for combinational and sequential circuits
12.	Compile and run SystemVerilog simulations
13.	Able to write SystemVerilog models for common digital circuits such as ALUs and finite state machines (FSMs)
14.	Able to interpret and analyze simulation results presented in both text logs and waveform displays
15.	Able to identify and resolve design bugs to ensure successful simulation
16.	Implement basic power-aware design techniques and understand fundamental Design for Test (DFT) concepts

11. Student Responsibilities:

Students must:

- Attend at least 80% of the course to be eligible for the final examination
- Engage in class discussion with respect and attention
- Self-study, direct their own studying—outside the classroom
- Complete all homework and assignments in a timely manner

12. Course assessment:

Assessment Components	Assessment types	Percentages (%)	Percentages of assessment	CLOs
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			components (%)	
A – On-going assessment	A1. Quiz	50	30	CLO 1-7
	A2. Homework	50		
B – Midterm exam	B1. Written test	100	20	CLO 1-6
C-Final exam	C1. Written test	100	20	CLO 1-7

D-Laboratory	D1. Correctness of design logic	40	30	CLO 8-11
	D2. Code structure	30		
	D3. Ability to trace and correct issues in simulation	30		

13. Lecture plan

Start date (proposed): September 3, 2025

Duration: 15 weeks

Weekly teaching: 2 sessions/week; session time: 10:00 am - 12:00 pm (VN time)

13.1 Lecture plan

	Session (2hrs)	Contents	Assessment Types	CLOs
Week 1 (01/9-07/9)	Session 1	Module 0: Setting the Stage: Designing the Un-Designable?	A1	CLO1
	Session 2	Module 1: Intro. to HDL/VDL & the IC Design/Verification Flow	A1, A2	CLO1,2
Week 2 (08/9-14/9)	Session 1	Module 2: SV Basic Syntax & Lexical Conventions 2.a: Key Concepts about Writing SV Code	A1, A2	CLO3
	Session 2	2.b SV Operators	A1, A2	
Week 3	Session 1	Module 3a: SV Data Type	A1, A2	CLO3
	Session 2	Module 3.b: SV Enhancements: Operators & Data Types Beyond Verilog	A1, A2	
Week 4	Session 1	Module 4a: Modules & Basic Structural Modeling	A1, A2	CLO4
	Session 2	Module 4b: Building Hierarchy	A1, A2	CLO4
Week 5	Session 1	Module 5: Procedural Blocks & Control Flow	A1, A2	CLO5
	Session 2	Module 5: Procedural Blocks & Control Flow (cont)	A1, A2	CLO5
Week 6	Session 1	Module 6a: Event-Driven Simulation	A1, A2	CLO6
	Session 2	Module 6b: Simulation Essential Topics	A1, A2	CLO6
Week 7	Session 1	Module 7a: Writing Tasks & Functions	A1, A2	CLO7

	Session 2	Module 7b: Using System Tasks & Functions	A1, A2	CLO7
Week 8	Session 1	Midterm Exam	B1	
	Session 2	Module 8 Standard Cell Library Overview	A1, A2	CLO8
Week 9	Session 1	Module 9a: Modeling Combinational and Sequential Logic Behavior	A1, A2	CLO9
	Session 2	Module 9b: Specify Blocks for Timing Information, Module 9c: Advanced Modeling Techniques	A1, A2	CLO9
Week 10	Session 1	Module 10: Hard Macro Modeling Concepts	A1, A2	CLO9
	Session 2	Module 11: Digital Logic Design Principle and RTL Design Styles	A1, A2	CLO10
Week 11	Session 1	Module 12: Synthesizable Logic Constructs: Combinational and Sequential	A1, A2	CLO11
	Session 2	Lab	D1, D2, D3	CLO12-15
Week 12	Session 1	Module 13: RTL Design for Common Digital Circuits	A1, A2	CLO11
	Session 2	Lab	D1, D2, D3	CLO12-15
Week 13	Session 1	Lab	D1, D2, D3	CLO12-15
	Session 2	Module 14: RTL Design for Finite State Machines (FSMs)	A1, A2	CLO11
Week 14	Session 1	Lab	D1, D2, D3	CLO12-15
	Session 2	Module 15: Beyond RTL: Power-Aware Design and DFT	A1, A2	CLO16
Week 15	Session 1	Final Exam	C1	

13.2. Lab plan

		Contents	Assessment Types	CLOs
Weeks 11	Session 2	Write RTL model of an ALU and its testbench, then simulate and verify functionality using text logs and waveform analysis.	D1, D2, D3	CLO12-15
Weeks 12	Session 2	Write RTL model of a single port RAM and its testbench, then simulate and verify functionality using text logs and waveform analysis.	D1, D2, D3	CLO12-15
Weeks 13	Session 1	Write RTL model of a FIFO and its testbench, then simulate and verify functionality using text logs and waveform analysis.	D1, D2, D3	CLO12-15
Weeks 14	Session 1	Write RTL model of a traffic light controller and its testbench, then simulate and verify functionality using text logs and waveform analysis.	D1, D2, D3	CLO12-15

14. Textbooks/References:

[1]. "SystemVerilog for Design" by Stuart Sutherland, Simon Davidmann, Peter Flake

[2]. SystemVerilog Language Reference Manual:

https://rfsoc.mit.edu/6S965/_static/F24/documentation/1800-2017.pdf